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PYTHON PROJECT REPORT

Forecasting Electricity Prices Using XGBoost

Major: Data Engineering and Software Development

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Executive Summary

This report presents a comprehensive machine learning solution developed for the purpose of forecasting electricity prices. Utilizing historical data on temperature and electricity demand, the project implements a sophisticated XGBoost regression model to predict electricity prices one time period ahead. The primary goal is to establish a robust baseline model that can serve as a foundation for future enhancements in the complex domain of energy market analytics.

The methodology encompasses a full data science pipeline, including data loading, exploratory analysis, rigorous preprocessing, feature engineering, model training, and evaluation. Key preprocessing steps involved the identification and removal of outliers using a 3-sigma statistical method and feature normalization via `StandardScaler`. The feature set was augmented with statistical indicators such as temporal means and trends to better capture underlying patterns.

The trained XGBoost model achieved an R^2 score of 0.4491 and a Root Mean Squared Error (RMSE) of 9.099 \$/MWh on the training dataset. On the unseen test dataset, the model yielded an R^2 score of 0.1464 and an RMSE of 19.895 \$/MWh. While the model demonstrates predictive capability, the performance gap between training and test sets indicates a degree of overfitting and highlights the challenges of generalizing to new data in a volatile market. The report concludes with a detailed analysis of these results, a discussion of the model's limitations, and a roadmap for future improvements.

Key Project Highlights

Aspect	Details
Dataset Size	941 training samples and 506 test samples after preprocessing.
Features	12 input features, including 6 original lagged variables (temperature and demand) and 6 engineered statistical features.
Model	XGBoost Regressor with `StandardScaler` for feature normalization.
Preprocessing	Outlier removal based on 3 standard deviations, feature scaling (Z-score), and statistical feature engineering.
Training Performance	RMSE: 9.099 \$/MWh; R ² Score: 0.4491
Test Performance	RMSE: 19.895 \$/MWh; R ² Score: 0.1464

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1.0 Introduction

1.1 Context

Electricity price forecasting (EPF) is a cornerstone of modern energy market operations and a subject of extensive research. The deregulation of electricity markets has transformed electrical energy into a commodity traded on various exchanges, leading to significant price volatility. This volatility is driven by a complex interplay of factors, including fuel costs, weather conditions, grid constraints, regulatory changes, and fluctuating consumer demand (Weron, 2014). Accurate price forecasts are therefore indispensable for a wide range of market participants.

The primary beneficiaries of precise EPF include:

- **Energy Traders and Producers:** Accurate forecasts enable the development of optimal bidding strategies in day-ahead and real-time markets, maximizing profits and minimizing financial risks.
- **Grid Operators:** System operators rely on price forecasts to anticipate periods of high demand and grid congestion, allowing for proactive management of supply resources to ensure grid stability and reliability.
- **Large Industrial Consumers:** Major energy consumers can schedule their high-load operations during periods of anticipated low prices, thereby reducing operational costs and improving their competitive edge.

- **Renewable Energy Providers:** For intermittent sources like solar and wind, price forecasts are crucial for planning energy storage and scheduling power delivery to the grid, optimizing revenue in a competitive environment.

Given the high-stakes nature of these decisions, the development of robust and reliable forecasting models is not merely an academic exercise but a critical business necessity. This project addresses this need by applying advanced machine learning techniques to historical market data.

1.2 Project Objectives

The principal objectives of this project were defined to ensure a structured and comprehensive approach to the forecasting problem. The key goals are as follows:

- To develop a predictive model capable of forecasting electricity prices for the next time period ($P(t+1)$) using historical temperature and demand data as inputs.
- To implement and document a robust data preprocessing pipeline, including techniques for handling outliers, scaling features, and engineering new variables to enhance model performance.
- To train and evaluate the selected model using standard industry metrics for regression tasks, such as Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2).
- To provide interpretable results, including feature importance analysis, that can offer actionable insights into the key drivers of electricity prices.

- To establish a well-documented baseline model that serves as a benchmark for future research, including explorations into more complex models, feature sets, and validation strategies.

1.3 Scope

The scope of this project is specifically defined to create a focused and achievable outcome within the given timeframe and resources. The project encompasses:

- **Time Series Analysis:** The core of the project involves analyzing time-ordered data to predict a future value based on past observations.
- **Supervised Machine Learning:** The problem is framed as a supervised regression task, where the model learns a mapping from input features to a continuous target variable.
- **Algorithm Focus:** The primary algorithm explored is the XGBoost Regressor, chosen for its proven performance in similar forecasting competitions and tasks.
- **Feature Engineering:** The project includes the creation of new features from the existing time series, specifically lagged variables and rolling statistical measures (mean, standard deviation, and trend).
- **Data Cleaning:** The methodology incorporates statistical preprocessing for outlier detection and removal to improve model robustness.
- **Evaluation and Visualization:** The final phase involves a quantitative evaluation of the model's predictive accuracy and the creation of visualizations to interpret its behavior and results.

2.0 Problem Statement

2.1 Research Question

The central research question guiding this project is: **Can we develop a machine learning model that accurately predicts the electricity price for the next time period, $P(t+1)$, using only historical temperature and demand data from the current and previous two periods (t , $t-1$, $t-2$)?**

2.2 Challenges

Forecasting electricity prices is an inherently difficult task due to several characteristics of the data and the underlying market dynamics. The key challenges addressed in this project are summarized in Table 2.

Challenge	Description
Non-linearity	The relationship between price and its drivers (like temperature and demand) is rarely linear. For instance, a small increase in temperature during a heatwave can cause a disproportionately large spike in demand and price. Linear models would fail to capture these complex interactions.
Volatility	Electricity prices are known for high volatility, including sudden and extreme price spikes. These events can be caused by unforeseen generator outages, transmission line failures, or sudden changes in weather, making the time series difficult to model.

Temporal Dependencies	As time series data, the observations are not independent. The price at time $t+1$ is strongly correlated with the price and conditions at time t . This temporal structure must be respected during model training and validation to avoid look-ahead bias (Hyndman & Athanasopoulos, 2018).
Outliers	The dataset contains extreme price events that are statistically significant outliers. These values can disproportionately influence the training of many models, leading to poor generalization on more typical data.
Feature Scaling	The input variables operate on vastly different scales. Temperature is measured in degrees Celsius (e.g., $\sim 25^{\circ}\text{C}$), while demand is in megawatts (e.g., ~ 5000 MW). Many algorithms, including those that use regularization, perform better when features are scaled to a similar range.

2.3 Success Criteria

The success of the project was evaluated against a set of predefined quantitative and qualitative criteria:

- **Predictive Accuracy:** The model must demonstrate a predictive power better than a naive baseline. A target R^2 score greater than 0.1 on the unseen test set was established as a minimum threshold for success.
- **Error Minimization:** The Root Mean Squared Error (RMSE) of the model should be reasonably low and, ideally, lower than that of basic statistical methods like a simple moving average.
- **Interpretability:** The model should provide insights into the factors driving its predictions. The feature importance scores generated by XGBoost are a key component of this criterion.
- **Computational Efficiency:** The model should be trainable and capable of making predictions within a reasonable timeframe, making it practical for potential real-world deployment.
- **Actionable Insights:** The results and analysis should provide clear, understandable insights that could inform the decisions of market stakeholders.

3.0 Dataset Description

3.1 Data Source

The data used for this project consists of time series information from the year 2018. The dataset was pre-divided into two separate files for training and testing purposes, ensuring that the test data remains unseen during the model development phase.

- **Training Data:** 2018_CI_Assignment_Training_Data.csv, containing an initial 956 samples.
- **Test Data:** 2018_CI_Assignment_Testing_Data.csv, containing 506 samples.

3.2 Features

The raw dataset contains seven columns, representing six input features and one target variable. The features are lagged measurements of temperature and electricity demand, designed to provide a temporal context for each prediction. The details of each feature are outlined in Table 3.

Feature	Description	Unit	Type
T(t-2)	Temperature two periods before the current time	°C	Input
T(t-1)	Temperature one period before the current time	°C	Input
T(t)	Temperature at the current time	°C	Input
D(t-2)	Electricity demand two periods before the current time	MW	Input
D(t-1)	Electricity demand one period before the current time	MW	Input
D(t)	Electricity demand at the current time	MW	Input
P(t+1)	Electricity price to be predicted for the next	\$/MWh	Target

	period		
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3.3 Statistical Summary

A preliminary statistical analysis of the training dataset provides crucial insights into the distribution and scale of each variable. Table 4 presents a summary of the key statistics for all features.

Statistic	T(t-2)	T(t-1)	T(t)	D(t-2)	D(t-1)	D(t)	P(t+1)
Count	956	956	956	956	956	956	956
Mean	26.60	26.60	26.60	5394.54	5394.84	5395.18	30.49
Std Dev	2.56	2.56	2.57	784.88	784.64	784.33	29.92
Min	20.90	20.90	20.90	3807.60	3807.60	3807.60	10.25
25%	24.60	24.60	24.60	4662.13	4664.48	4668.20	18.99
50%	26.20	26.20	26.20	5396.60	5396.60	5396.60	24.80
75%	28.80	28.80	28.80	6113.90	6113.90	6113.90	32.13
Max	32.30	32.30	32.30	6870.00	6870.00	6870.00	460.29

3.4 Data Quality Assessment

Based on the initial data loading and statistical summary, the following observations were made regarding data quality:

- **Completeness:** An initial check confirmed that there are no missing or null values in either the training or test datasets, simplifying the preprocessing pipeline.
- **Data Types:** All columns are of a numerical data type, suitable for direct input into machine learning algorithms after appropriate scaling.

- **Price Volatility:** The target variable, $P(t+1)$, exhibits high volatility. Its standard deviation (29.92) is nearly as large as its mean (30.49), indicating a wide distribution of price values.
- **Outlier Presence:** The maximum price of 460.29 \$/MWh is an order of magnitude greater than the 75th percentile (32.13 \$/MWh). This strongly suggests the presence of extreme price spikes that are statistical outliers and require special handling.
- **Feature Consistency:** The statistical properties (mean, std dev, etc.) of the lagged temperature and demand features are highly consistent with each other, which is expected for consecutive time series measurements.

4.0 Methodology

4.1 Overall Approach

The project was executed following a standard, structured machine learning workflow. This systematic approach ensures reproducibility and allows for clear documentation at each stage.

The pipeline can be summarized as follows:

Data Loading → Exploratory Data Analysis (EDA) → Data Preprocessing → Feature Engineering
→ Model Training → Prediction on Test Data → Performance Evaluation → Visualization and Interpretation

Each step was implemented sequentially, with insights from earlier stages informing the decisions made in later ones. For example, the discovery of outliers during EDA directly led to the implementation of an outlier removal step in the preprocessing phase.

4.2 Algorithm Selection: XGBoost

The choice of the predictive algorithm is a critical decision in any machine learning project. For this task, the **XGBoost (Extreme Gradient Boosting)** algorithm was selected. XGBoost is an optimized and scalable implementation of gradient boosted decision trees, and it has become a de facto standard for winning competitions on structured or tabular data (Chen & Guestrin, 2016). The rationale for its selection is based on several key advantages, detailed in Table 5.

Advantage	Description
Predictive Power	XGBoost builds a model in the form of an ensemble of weak prediction models, typically decision trees. It adds new trees sequentially to correct the errors made by previous ones, resulting in a highly accurate and robust final model.
Handles Non-linearity	As a tree-based model, XGBoost is inherently capable of capturing complex, non-linear relationships and interactions between features without requiring manual transformations. This is ideal for modeling the complex dynamics of electricity markets.
Robustness to Outliers	While sensitive to outliers, decision tree

	ensembles are generally more robust to them than linear models. The boosting process can learn to isolate the effect of extreme values in specific tree branches.
Built-in Regularization	XGBoost includes L1 (Lasso) and L2 (Ridge) regularization terms in its objective function. This helps to control the complexity of the model and prevent overfitting, which is a common problem in time series forecasting.
Feature Importance	The algorithm provides scores that rank the relative importance of each feature in the construction of the model's predictions. This is invaluable for model interpretability and for understanding the underlying drivers of the target variable.

4.3 Data Preprocessing Techniques

Raw data is rarely suitable for direct use in machine learning models. The preprocessing phase is designed to clean, transform, and enrich the data to improve model performance and reliability.

4.3.1 Outlier Detection and Removal

As identified in the EDA phase, the target variable $P(t+1)$ contained significant outliers. To mitigate their influence, an outlier removal strategy was implemented based on the standard deviation method.

- **Method:** The 3-sigma rule was applied, which defines an outlier as any data point that falls outside of three standard deviations from the mean.
- **Calculation:** The upper threshold was calculated as: $\text{Threshold} = \mu + 3 * \sigma$, where μ is the mean and σ is the standard deviation of $P(t+1)$. This resulted in a price threshold of **120.25 \$/MWh**.
- **Result:** Any sample in the training data with a price exceeding this threshold was removed. This led to the removal of **15 outlier samples**, reducing the training set size from 956 to 941. This step helps prevent the model from being skewed by rare, extreme events.

4.3.2 Feature Engineering

To provide the model with more information than just the raw lagged values, several new features were engineered based on the existing data. These statistical features aim to summarize the recent state and trend of the system.

- **Temporal Means:** T_mean and D_mean were created by calculating the average of the temperature and demand values across the available time lags ($t, t-1, t-2$). These features represent the recent average condition.
- **Temporal Standard Deviations:** T_std and D_std were calculated to represent the recent volatility or stability of temperature and demand.

- **Trend Features:** ``T_trend`` and ``D_trend`` were calculated as the difference between the current value and the value two periods ago (e.g., ``D(t) - D(t-2)``). These features capture the short-term direction and magnitude of change.

This process expanded the feature space from 6 to 12 input variables.

4.3.3 Feature Scaling

To ensure that all features contribute fairly to the model and to aid the optimization process, feature scaling was applied.

- **Method:** The ``StandardScaler`` from the scikit-learn library was used (Pedregosa et al., 2011). This method performs Z-score normalization.
- **Transformation:** Each feature ``X`` is transformed by subtracting its mean ``μ`` and dividing by its standard deviation ``σ``: $X_{\text{scaled}} = (X - \mu) / \sigma$.
- **Implementation:** The scaler was fitted **only** on the training data to prevent data leakage from the test set. The same fitted scaler was then used to transform both the training and test data, ensuring consistency.

5.0 Implementation Details

5.1 Technology Stack

The project was developed entirely in the Python ecosystem, leveraging a suite of open-source libraries that are standard for data science and machine learning tasks. The key technologies are listed in Table 6.

Component	Technology/Library	Version	Purpose
Programming Language	Python	3.8+	Core language for all scripting and logic.
Data Manipulation	pandas	Latest	Data loading, cleaning, transformation, and analysis.
Machine Learning	XGBoost	Latest	Implementation of the gradient boosting model.
Preprocessing	scikit-learn	0.24.2+	Used for `StandardScaler` and evaluation metrics.
Visualization	Matplotlib, Seaborn	Latest	Generating plots and charts for analysis and reporting.

5.2 Key Implementation Components

The core of the implementation is the configuration and training of the XGBoost model. A set of hyperparameters was chosen to provide a solid baseline. These parameters control the model's

learning process and complexity. While not exhaustively tuned, they were selected based on common practices to balance performance and prevent overfitting. The model was instantiated with the following configuration:

```
# XGBoost model configuration

xgb_model = XGBRegressor(

    n_estimators=100,    # Number of boosting rounds (trees)

    learning_rate=0.01,  # Step size shrinkage to prevent overfitting

    max_depth=4,        # Maximum depth of a tree

    subsample=0.7,      # Fraction of samples to be used for fitting each tree

    colsample_bytree=0.7, # Fraction of features to be used for fitting each tree

    reg_alpha=1.0,      # L1 regularization term on weights

    reg_lambda=1.0,     # L2 regularization term on weights

    min_child_weight=5,  # Minimum sum of instance weight needed in a child

    random_state=42     # Seed for reproducibility

)
```

This configuration defines a moderately complex model with a slow learning rate and strong regularization (`reg_alpha`, `reg_lambda`) to promote generalization. The `subsample` and `colsample_bytree` parameters introduce stochasticity, which further helps in reducing overfitting.

6.0 Results and Analysis

6.1 Performance Metrics

The performance of the trained XGBoost model was evaluated on both the training data (to assess fit) and the unseen test data (to assess generalization). The primary metrics used were Root Mean Squared Error (RMSE) and the Coefficient of Determination (R^2). The results are summarized in Table 7.

Dataset	RMSE (\$/MWh)	R^2 Score
Training	9.0992	0.4491
Test	19.8953	0.1464

Analysis of Metrics:

- RMSE:** The RMSE on the training set (9.10 \$/MWh) is significantly lower than on the test set (19.90 \$/MWh). This indicates that the model's predictions are, on average, about twice as far from the actual values on new data as they are on the data it was trained on. This discrepancy is a classic sign of overfitting.
- R^2 Score:** The R^2 score of 0.4491 on the training data suggests that the model can explain approximately 45% of the variance in the training prices. However, this explanatory power drops to just 14.6% on the test data. While the test R^2 of 0.1464 is positive (indicating the model is better than a naive model that always predicts the mean), it is quite low, confirming that the model's ability to generalize is limited.

The significant gap between training and test performance is the most critical finding. It suggests that the model has learned patterns specific to the training data that do not hold true for the test data.

6.2 Feature Importance

XGBoost provides a mechanism to evaluate the importance of each feature in the construction of the model. This is crucial for understanding what factors the model considers most predictive. The most important features, as identified by the model, are:

1. D_mean (Average demand over recent periods)
2. $D(t)$ (Current electricity demand)
3. D_trend (Recent trend in demand)
4. T_mean (Average temperature over recent periods)
5. $D(t-1)$ (Demand from the previous period)

Analysis of Importance: It is evident that demand-related features are the dominant drivers of the model's predictions. The engineered feature ``D_mean`` is ranked as the most important, highlighting the value of feature engineering. The current demand ``D(t)`` and the recent trend ``D_trend`` are also highly influential. This aligns with economic principles, where demand is a primary determinant of price. Temperature features, while present, are ranked lower, suggesting they are secondary drivers in this model's view, likely influencing price indirectly through their effect on demand.

7.0 Visualizations

A series of visualizations were planned to complement the quantitative analysis and provide a more intuitive understanding of the data and model performance. While the plots themselves are generated in the accompanying notebook, their purpose and expected insights are described here.

- **Price Distribution Analysis (Histogram with KDE):** This plot would visualize the distribution of the target variable, $P(t+1)$. It would clearly show the right-skewed nature of the data and the presence of a long tail, confirming the existence of price spikes and justifying the outlier removal step.
 - **Correlation Heatmap:** A heatmap of the correlation matrix for all features would reveal the linear relationships between variables. It would be expected to show strong positive correlations among the lagged demand features and among the lagged temperature features, as well as a notable positive correlation between demand features and the price.
 - **Actual vs. Predicted Scatter Plot:** This is a key diagnostic plot. It would scatter the actual prices from the test set against the model's predicted prices. For a perfect model, all points would lie on a 45-degree line. For this model, we would expect to see a cloud of points with a positive correlation but significant spread, visually representing the R^2 score of 0.1464.
 - **Time Series Prediction Plot:** This plot would overlay the actual price time series from the test set with the predicted time series. It would help to visualize where the model succeeds and fails. We would likely see that the model captures the general trend but fails to predict the magnitude and timing of sharp price spikes.
 - **Residual Analysis (Histogram or Scatter Plot):** A plot of the residuals (prediction errors) would help diagnose model bias. An ideal model would have residuals centered around zero and randomly distributed. Any discernible pattern in the residuals would suggest that the model is failing to capture some underlying structure in the data.
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8.0 Discussion

8.1 Strengths of the Approach

Despite the limitations in performance, the methodology employed in this project has several notable strengths that establish a solid foundation for future work.

- **Solid Foundation:** The project successfully implements a complete, end-to-end machine learning pipeline, from data ingestion to evaluation. This structure is modular and can be easily adapted for future experiments.
- **Performant Model Choice:** The selection of XGBoost was appropriate for the task, as it is well-suited to capturing the non-linearities inherent in electricity price data.
- **Systematic Preprocessing:** The approach to data cleaning and feature engineering was systematic and well-justified. The removal of outliers and scaling of features are crucial steps that improve the reliability of the modeling process.
- **Reproducibility:** By documenting the methodology, technology stack, and model configuration (including the `random_state`), the project ensures that its results are reproducible, which is a cornerstone of good scientific practice.

8.2 Limitations and Challenges

The primary challenge revealed by the results is the model's limited ability to generalize to unseen data. This and other limitations are discussed below.

- **Generalization Gap (Overfitting):** The large discrepancy between training ($R^2=0.45$) and test ($R^2=0.15$) performance is a clear indication of overfitting. The model has learned noise and

specific patterns in the training set that do not apply to the test set. This could be due to the model being too complex for the amount of data available or the inherent differences between the training and testing periods.

- **Limited Feature Scope:** The model was constrained to using only historical temperature and demand data. Real-world electricity prices are influenced by a much wider range of factors, including fuel prices (e.g., natural gas), renewable energy generation levels, grid congestion, and day-of-week/time-of-day effects (Weron, 2014). The absence of these features limits the model's potential accuracy.
- **Simple Temporal Dynamics:** The use of simple lagged features and rolling statistics may not be sufficient to capture the complex temporal dynamics, seasonality, and auto-correlation present in energy price series. More sophisticated time series techniques may be required.
- **Static Validation:** The use of a single train-test split is standard but can be sensitive to the specific period chosen for the split. The performance might be different if another period was used for testing.

9 Conclusions

9.1 Results Summary

This project successfully developed and evaluated an XGBoost-based model for forecasting electricity prices. The key conclusions drawn from the results are:

- An end-to-end machine learning pipeline was successfully implemented, demonstrating a complete workflow for tackling a time series forecasting problem.
- A baseline performance was established, with the model achieving an R^2 score of **0.1464** and an RMSE of **19.895 \$/MWh** on the unseen test set. This confirms that the model has some predictive power, albeit limited.
- A significant performance gap between the training and test data was observed, highlighting a clear problem with model generalization and overfitting.
- Feature importance analysis revealed that demand-related variables, particularly engineered features like the moving average of demand, are the most significant predictors in the current model.

9.2 Future Improvements

The baseline model developed in this project opens several avenues for future improvement. The following steps are recommended to enhance the model's predictive accuracy and robustness:

- **Advanced Feature Engineering:** Incorporate more domain-specific features. This could include cyclical features (e.g., time of day, day of week, month of year), interaction terms (e.g., temperature * demand), and potentially integrating external data sources like fuel prices or weather forecasts.

- **Hyperparameter Tuning:** Conduct a systematic hyperparameter optimization for the XGBoost model. Techniques like Grid Search, Randomized Search, or Bayesian Optimization could be used to find a set of parameters that improves generalization and reduces overfitting.
- **Enhanced Validation Strategy:** Replace the simple train-test split with a more robust time-series cross-validation technique, such as `TimeSeriesSplit` in scikit-learn or a forward-chaining approach. This provides a more reliable estimate of the model's performance on unseen data (Hyndman & Athanasopoulos, 2018).
- **Exploration of Alternative Models:** While XGBoost is powerful, other models could be explored. This includes other ensemble methods like LightGBM or CatBoost, as well as models specifically designed for time series, such as LSTMs (Long Short-Term Memory networks) or Transformers, which may be better at capturing long-term dependencies.

10.0 Appendices

Appendix A: Dataset Files

The project utilized the following data files:

- 2018_CI_Assignment_Training_Data.csv
- 2018_CI_Assignment_Testing_Data.csv

Appendix B: Evaluation Metrics

Root Mean Squared Error (RMSE): Measures the square root of the average of the squared differences between predicted and actual values. It is sensitive to large errors and is in the same units as the target variable.

Formula: $RMSE = \sqrt{[\sum (y_i - \hat{y}_i)^2 / n]}$

Coefficient of Determination (R²): Represents the proportion of the variance in the dependent variable that is predictable from the independent variable(s). An R² of 1 indicates perfect prediction, while an R² of 0 indicates the model is no better than predicting the mean.

Formula: $R^2 = 1 - [SS_{res} / SS_{tot}]$

Appendix C: Libraries Used

The core Python libraries used in this project include: pandas, numpy, scikit-learn, xgboost, matplotlib, and seaborn.

Appendix D: Complete Results

Detailed results, including per-sample predictions and residuals, are available in the output files generated by the project's Jupyter Notebook.

Appendix E: Notebook Organization

The accompanying Jupyter Notebook is structured sequentially, following the methodology outlined in this report. Sections include Data Loading, EDA, Preprocessing, Model Training, Evaluation, and Visualization.

Appendix F: References

- [1] Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining* (pp. 785–794). ACM.
- [2] Hyndman, R. J., & Athanasopoulos, G. (2018). *Forecasting: principles and practice* (2nd ed.). OTexts. <https://otexts.com/fpp2/>
- [3] Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., ... & Duchesnay, E. (2011). Scikit-learn: Machine Learning in Python. *Journal of Machine Learning Research*, 12, 2825-2830.
- [4] Weron, R. (2014). Electricity price forecasting: A review of the state-of-the-art with a look into the future. *International Journal of Forecasting*, 30(4), 1030-1081.